Load Balancer & Auto Scaling- Question Bank

A. Conceptual Questions

1. What is a Load Balancer in AWS, and why is it used?
2. What is the main purpose of distributing traffic across multiple EC2 instances?
3. What is the difference between **ALB, NLB, and CLB**?
4. What is the difference between a **Load Balancer** and an **Auto Scaling Group**?
5. What is a Target Group, and why is it required?
6. What is a Listener in a Load Balancer?
7. What is a Listener Rule in an ALB?
8. What is a Health Check, and why is it important?
9. What happens when an EC2 instance fails the Load Balancer health check?
10. What is the difference between **high availability** and **load balancing**?
11. Why should Load Balancers generally be deployed across multiple Availability Zones?
12. What is the difference between **Layer 4** and **Layer 7** load balancing?
13. Which AWS Load Balancer works at Layer 4, and which works at Layer 7?
14. What is the role of DNS in an AWS Load Balancer?

B. Technical Questions

15. What is the difference between **ALB Listener on port 80** and **Target Group port 80**?
16. What is the difference between an **ALB Security Group** and an **EC2 Security Group**?
17. What inbound rules should normally be configured on the ALB Security Group and EC2 Security Group?
18. How does an ALB decide which target should receive an HTTP request?
19. What is **path-based routing** in ALB? Give an example.

20. What is **host-based routing** in ALB? Give an example.
21. What is the difference between **Internet-facing** and **Internal** Load Balancers?
22. What is a **Network Load Balancer**, and why is it suitable for TCP/UDP traffic?
23. What is the difference between an **NLB listener** and an **ALB listener**?
24. What is **Classic Load Balancer**, and how is it different from ALB and NLB?
25. What is a **Launch Template**, and why is it commonly used with Auto Scaling?
26. What is the difference between **Minimum**, **Desired**, and **Maximum capacity** in an Auto Scaling Group?
27. What is **Dynamic Scaling**, and how can CPU utilization trigger Auto Scaling?
28. What is the purpose of an **Auto Scaling Policy**?
29. What happens when the desired capacity of an Auto Scaling Group is increased from 2 to 5?
30. What happens to the Load Balancer's Target Group when Auto Scaling launches a new EC2 instance?

C. Scenario-Based Questions

31. You have **two EC2 instances in different Availability Zones**, but the ALB shows only one instance as healthy. What would you check?
32. Users can access the ALB DNS name, but they receive **502 Bad Gateway**. What could be the possible causes?
33. Your ALB is active and both EC2 instances are running, but the websites are not opening through the ALB DNS name. What would you troubleshoot?
34. Your ALB health check is configured for /, but your application actually responds successfully only at /health. What should you change?
35. An application receives normal traffic during the day and very high traffic during the evening. How would you configure Auto Scaling to handle this change?

36. An application currently has **2 EC2 instances**. CPU utilization reaches 80%, and you want Auto Scaling to launch additional instances. What configuration would you create?
37. An EC2 instance becomes unhealthy while it is registered with an ALB. What will happen to traffic going to that instance?
38. You need to expose a web application using HTTP/HTTPS and route /login and /payment to different groups of servers. Which Load Balancer would you consider, and how would you configure it?
39. Your application uses **TCP traffic** and requires very high throughput and low latency. Which Load Balancer would you consider, and why?
40. You have an ALB + Auto Scaling architecture with **2 EC2 instances initially**. Traffic increases suddenly, Auto Scaling launches 3 additional instances, and all 5 instances become healthy. Explain the complete traffic flow from the user to the ALB, Target Group, and EC2 instances.

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